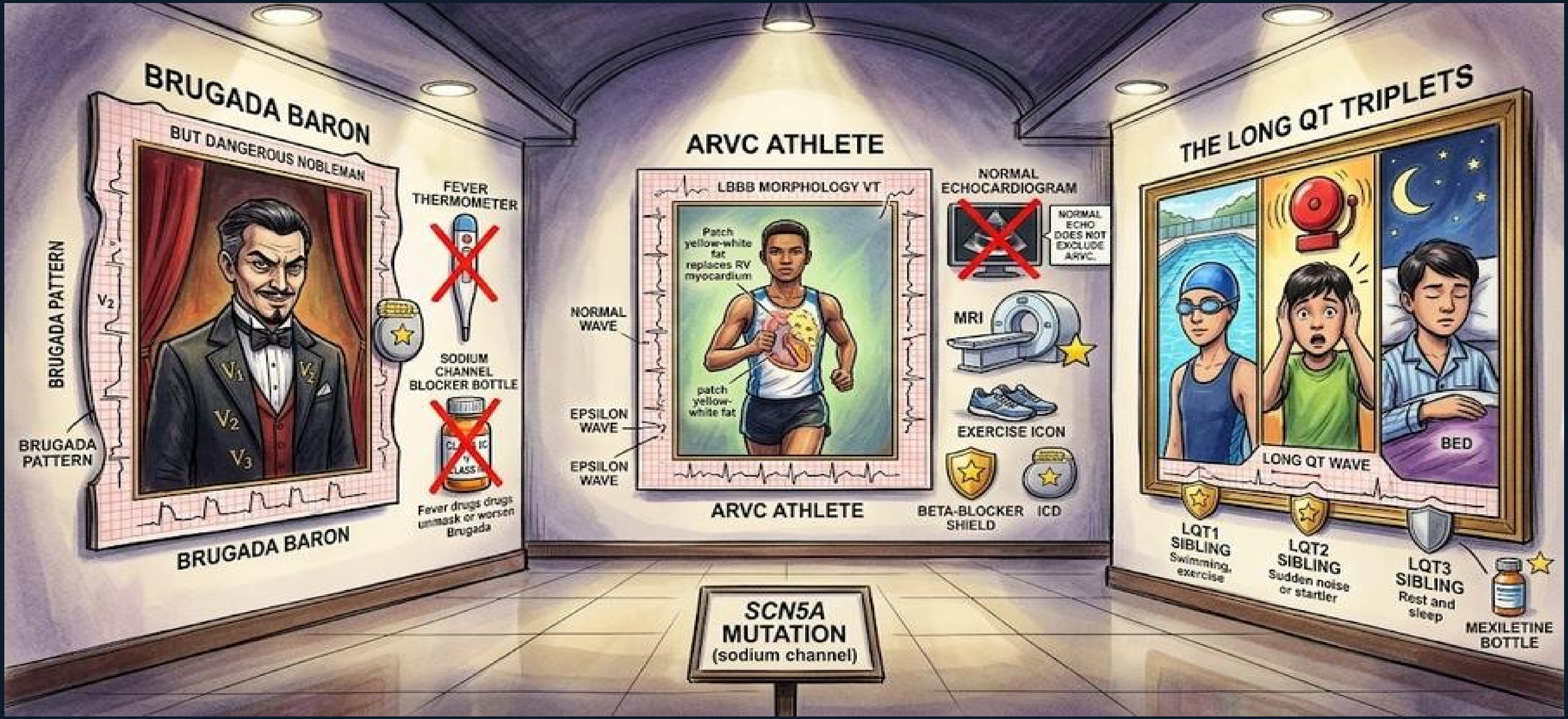
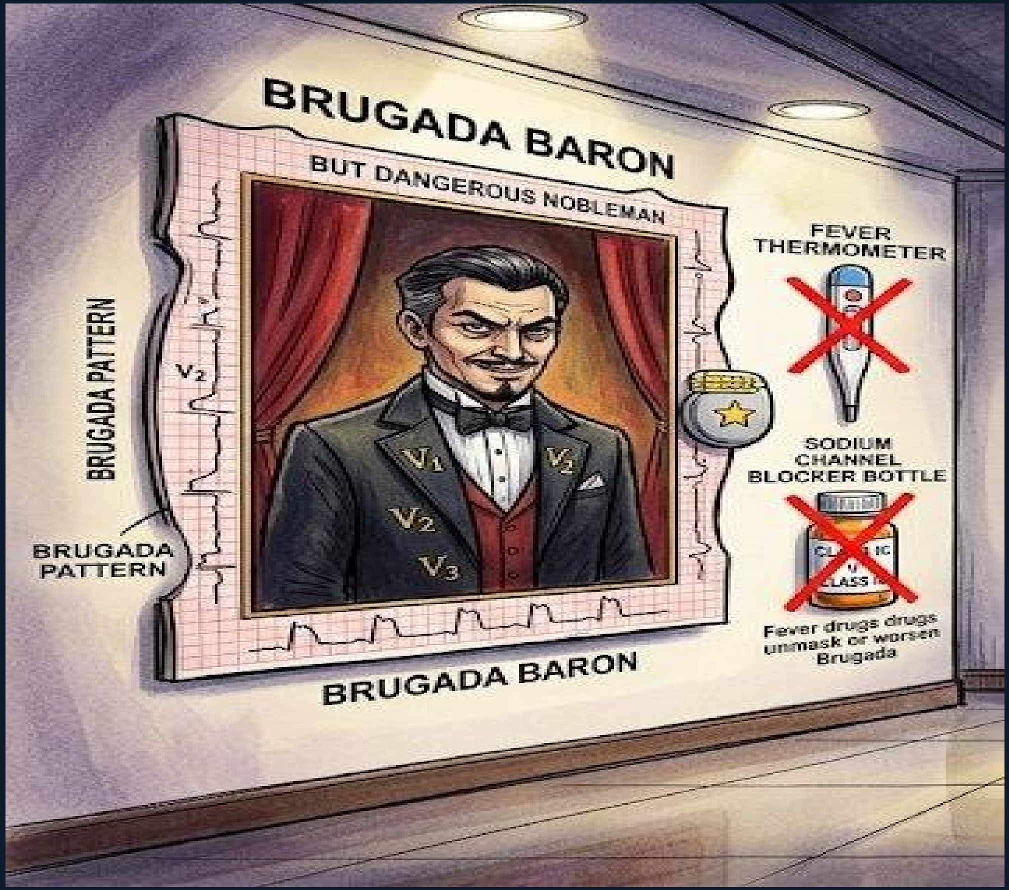
Scene G: Channelopathies & Inherited Arrhythmia Syndromes



Genetics Museum · IMPORTANT · 2 Board Questions

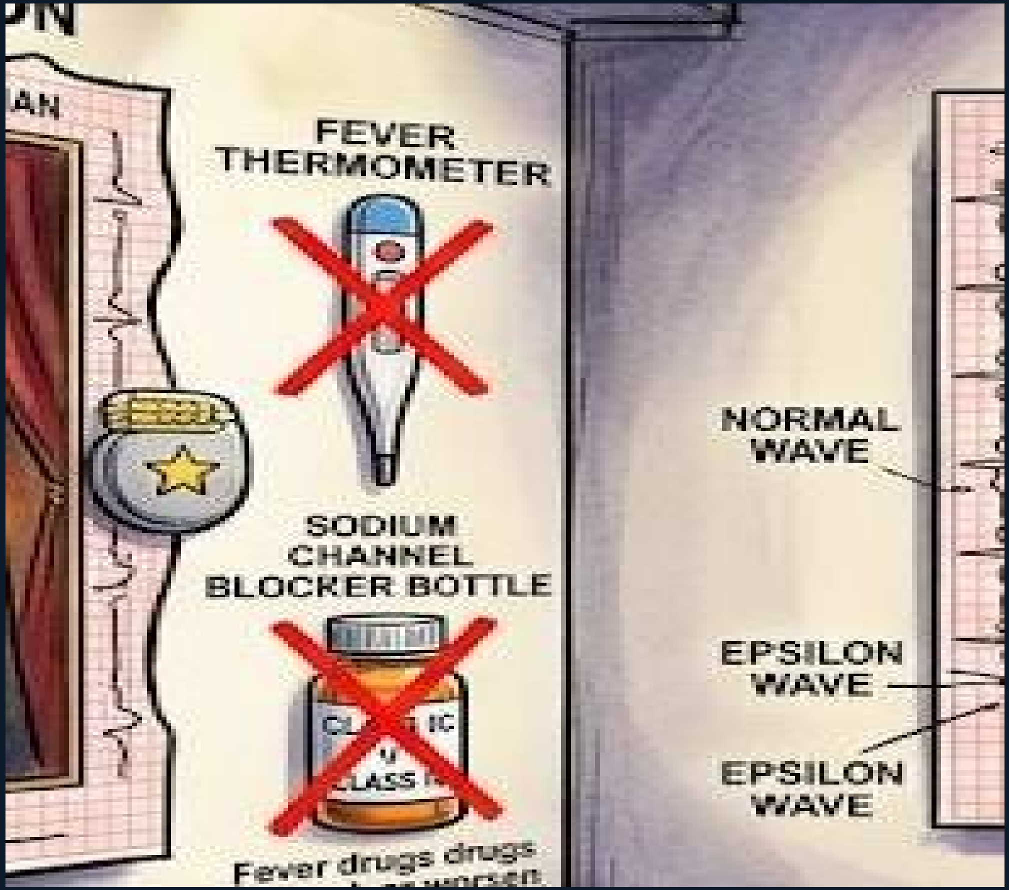


Brugada Baron — Portrait 1

The Brugada Baron portrait is framed by an ECG strip showing the classic pattern: coved (tombstone-shaped) ST elevation in V1, V2, V3 with RBBB morphology. A sinister nighttime expression encodes sudden death at rest or sleep.

Brugada syndrome: sodium channel mutation (SCN5A). ECG shows coved ST elevation ≥2 mm in ≥1 of V1–V3 with RBBB-like morphology (Type 1 pattern). Spontaneous or drug-induced.

Classic presentation: young Southeast Asian male, sudden death at rest/sleep, no structural heart disease on echo.

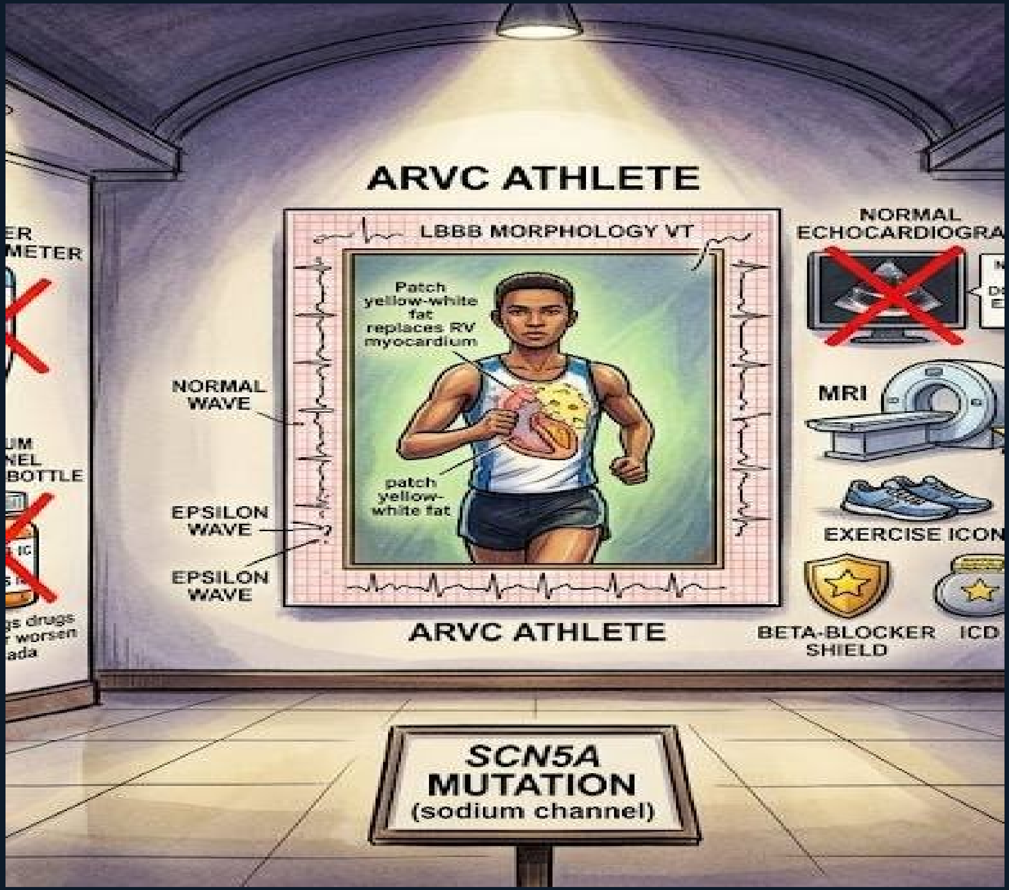


Brugada — Triggers and Treatment

A red X on a fever thermometer and a red X on a sodium channel blocker bottle (Class IC/IA) flank the Brugada portrait. A gold-star ICD is mounted on the frame.

Fever is a major trigger for Brugada arrhythmias — treat aggressively with antipyretics. Sodium channel blockers (class IC: flecainide, propafenone; class IA: procainamide) can unmask or worsen Brugada pattern.

Treatment: ICD is the definitive treatment for symptomatic Brugada. Quinidine may suppress arrhythmias pharmacologically. Board trap: 'a patient with Brugada pattern started on flecainide for AF' — wrong; sodium channel blockers are contraindicated.



ARVC Athlete — Portrait 2

A young runner's RV shows patches of yellow-white fat replacing the myocardium (Swiss cheese appearance). The ECG strip framing shows LBBB morphology VT and an epsilon wave — a small deflection after the QRS.

ARVC: fatty/fibrous replacement of RV myocardium. Exercise-triggered VT or VF in young athletes. ECG: LBBB morphology VT (RV origin → activation goes left → LBBB pattern), epsilon wave in V1–V3, T-wave inversions V1–V3.

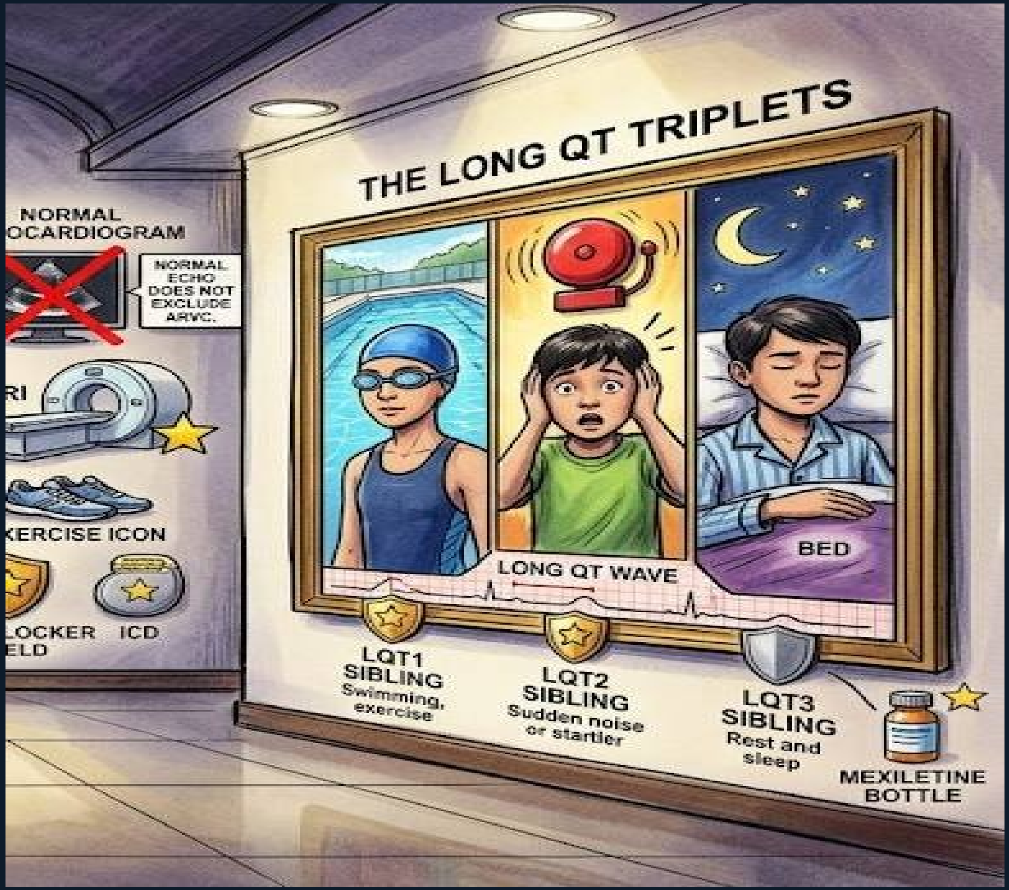
Diagnosis: Task Force criteria. Cardiac MRI is superior to echo for detecting fatty infiltration and fibrous replacement.

ARVC — Echo Trap and MRI Gold Star

A normal echocardiogram screen has a large red X with the label 'NORMAL ECHO DOES NOT EXCLUDE ARVC.' An MRI machine beside it has a gold star.

Early ARVC can have a completely normal echocardiogram. Cardiac MRI detects fatty infiltration and late gadolinium enhancement in the RV free wall — it is the superior diagnostic tool.

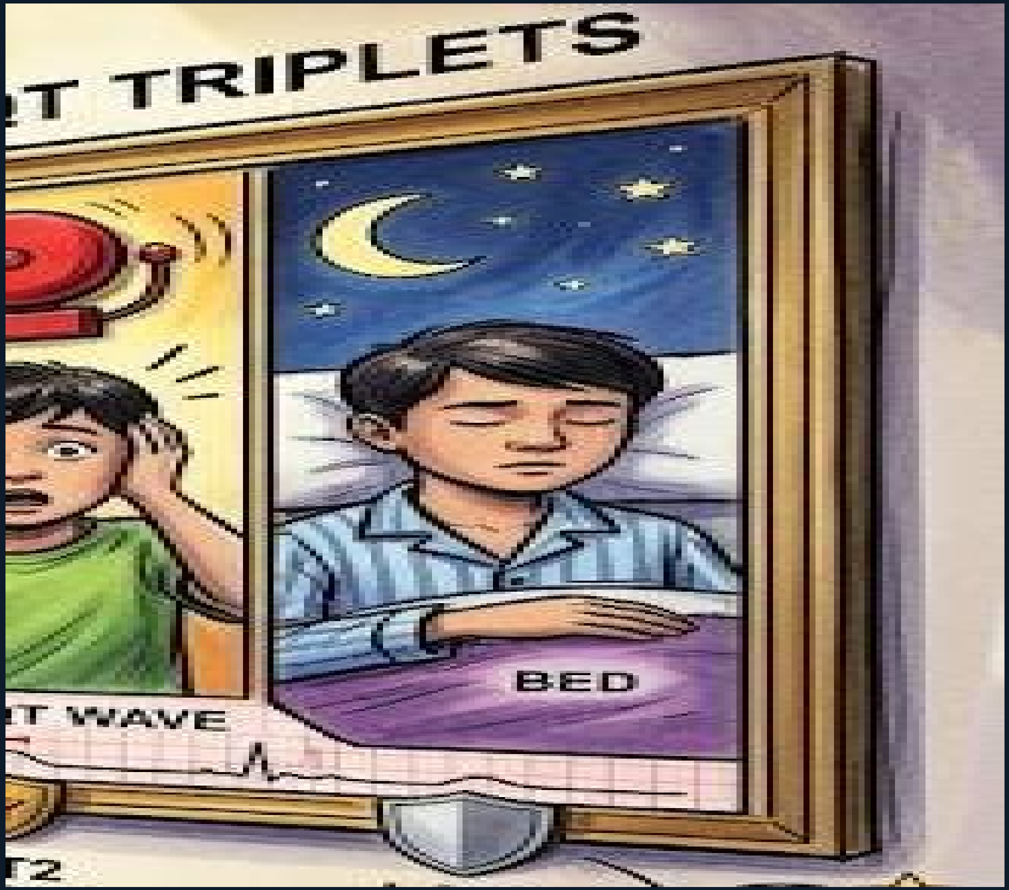
Board trap: 'echo was normal, so ARVC is ruled out' — wrong. MRI is required when ARVC is suspected. Running shoes in the scene remind you: exercise-triggered arrhythmia in a young athlete with LBBB VT = think ARVC until proven otherwise.



The Long QT Triplets — LQT1, 2, 3

Three siblings in one portrait frame. LQT1: swimming cap (trigger = exercise/swimming). LQT2: covering ears from alarm bell (trigger = sudden noise/startle). LQT3: pajamas in bed with moon (trigger = rest/sleep).

Congenital long QT syndromes are caused by mutations in cardiac ion channels. LQT1 (KCNQ1 = Iks channel): beta-blockers highly effective. LQT2 (KCNH2 = IKr channel): avoid loud noises; beta-blockers partially effective. LQT3 (SCN5A = sodium channel): occurs at rest/sleep; beta-blockers less effective.



LQT3 — Different Treatment

LQT3 sibling in pajamas with a slightly dimmer beta-blocker shield beside a mexiletine bottle.

LQT3 results from a gain-of-function mutation in SCN5A — persistent sodium current during repolarization. Beta-blockers are less effective here than in LQT1 or LQT2.

Treatment: ICD is recommended for symptomatic LQT3. Mexiletine (class IB — shortens QT by blocking the persistent Na current) can be added. Board trap: 'beta-blockers are equally effective in all congenital long QT subtypes' — wrong. LQT3 is the exception.

Scene G — Board Trap Master Summary

Brugada: coved ST elevation V1–V3 + RBBB

SCN5A mutation; spontaneous or drug-induced

Fever triggers Brugada

Treat aggressively; maintain normothermia

Na channel blockers CI in Brugada

Class IC/IA unmask or worsen pattern

ARVC: LBBB morphology VT (not RBBB)

RV origin → activation goes left → LBBB

Epsilon wave = ARVC hallmark

Small deflection after QRS in V1–V3

Normal echo does NOT exclude ARVC

Use cardiac MRI for diagnosis

LQT1: exercise/swimming trigger

Beta-blockers highly effective

LQT2: sudden noise/startle trigger

Avoid loud noises; partial beta-blocker response

LQT3: rest/sleep trigger

Beta-blockers less effective; add mexiletine; ICD

SCN5A = both Brugada and LQT3

Loss of function in Brugada; gain in LQT3